

The Cobb-Douglas Production Model

In 1928, Cobb and Douglas used a similar function to model the production of the entire US economy in the first quarter of this century. Using government estimates of P , the total yearly production between 1899 and 1922, of K , the total capital investment over the same period, and of L , the total labor force, they found that P was well approximated by the *Cobb-Douglas production function*

$$P = 1.01L^{0.75}K^{0.25}.$$

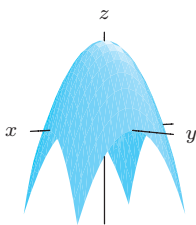
This function turned out to model the US economy surprisingly well, both for the period on which it was based, and for some time afterward.

Exercises and Problems for Section 12.3

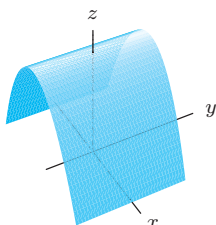
Exercises

In Exercises 1–4, sketch a possible contour diagram for each surface, marked with reasonable z -values. (Note: There are many possible answers.)

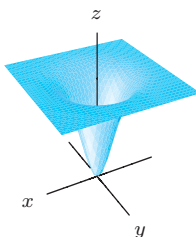
1.



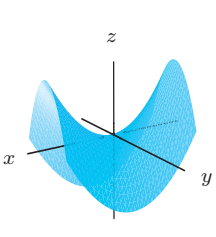
2.



3.



4.



14. Let $f(x, y) = 3x^2y + 7x + 20$. Find an equation for the contour that goes through the point $(5, 10)$.

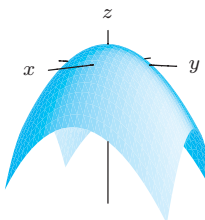
15. (a) For $z = f(x, y) = xy$, sketch and label the level curves $z = \pm 1$, $z = \pm 2$.

(b) Sketch and label cross-sections of f with $x = \pm 1$, $x = \pm 2$.

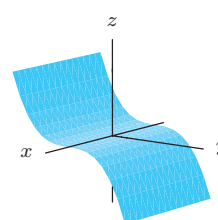
(c) The surface $z = xy$ is cut by a vertical plane containing the line $y = x$. Sketch the cross-section.

16. Match the surfaces (a)–(e) in Figure 12.48 with the contour diagrams (I)–(V) in Figure 12.49.

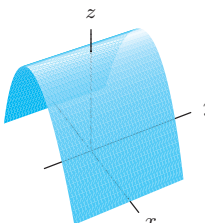
(a)



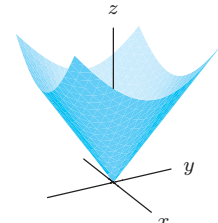
(b)



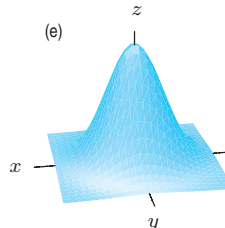
(c)



(d)



(e)



In Exercises 5–13, sketch a contour diagram for the function with at least four labeled contours. Describe in words the contours and how they are spaced.

5. $f(x, y) = x + y$

6. $f(x, y) = 3x + 3y$

7. $f(x, y) = x^2 + y^2$

8. $f(x, y) = -x^2 - y^2 + 1$

9. $f(x, y) = xy$

10. $f(x, y) = y - x^2$

11. $f(x, y) = x^2 + 2y^2$

12. $f(x, y) = \sqrt{x^2 + 2y^2}$

13. $f(x, y) = \cos \sqrt{x^2 + y^2}$

Figure 12.48

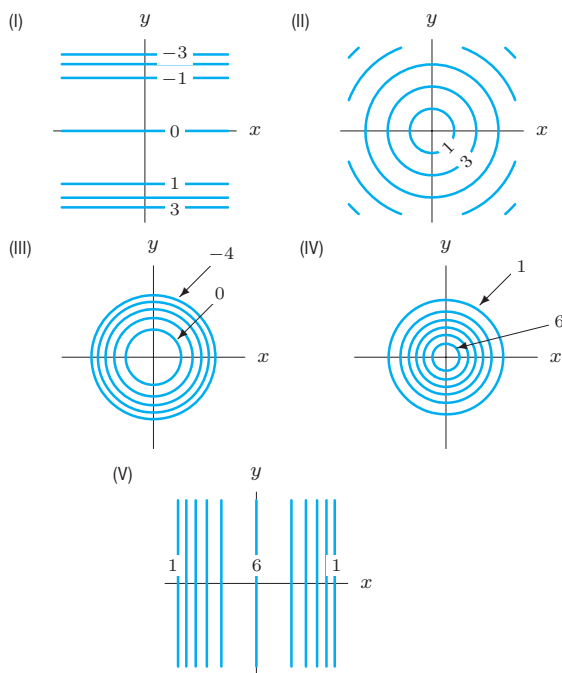


Figure 12.49

Table 12.7

$y \backslash x$	-1	0	1
-1	2	0	2
0	2	0	2
1	2	0	2

Table 12.8

$y \backslash x$	-1	0	1
-1	2	2	2
0	0	0	0
1	2	2	2

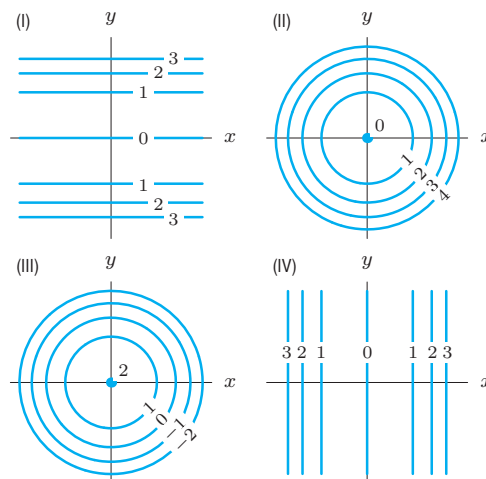


Figure 12.50

17. Match Tables 12.5–12.8 with contour diagrams (I)–(IV) in Figure 12.50.

Table 12.5

$y \backslash x$	-1	0	1
-1	2	1	2
0	1	0	1
1	2	1	2

Table 12.6

$y \backslash x$	-1	0	1
-1	0	1	0
0	1	2	1
1	0	1	0

18. Figure 12.51 shows a graph of $f(x, y) = (\sin x)(\cos y)$ for $-2\pi \leq x \leq 2\pi$, $-2\pi \leq y \leq 2\pi$. Use the surface $z = 1/2$ to sketch the contour $f(x, y) = 1/2$.

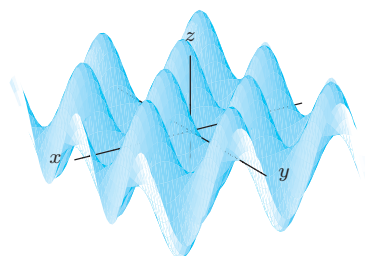


Figure 12.51

Problems

19. Total sales, Q , of a product are a function of its price and the amount spent on advertising. Figure 12.52 shows a contour diagram for total sales. Which axis corresponds to the price of the product and which to the amount spent on advertising? Explain.

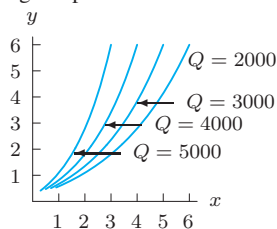


Figure 12.52

20. Figure 12.53 shows contours of $f(x, y) = 100e^x - 50y^2$. Find the values of f on the contours. They are equally spaced multiples of 10.

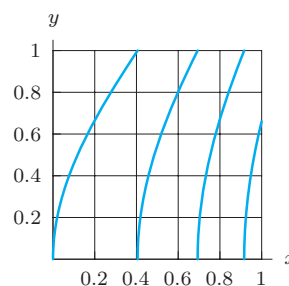


Figure 12.53

21. Match the functions (a)–(f) with the level curves (I)–(VI):

- (a) $f(x, y) = x^2 - y^2 - 2x + 4y - 3$
- (b) $g(x, y) = x^2 + y^2 - 2x - 4y + 15$
- (c) $h(x, y) = -x^2 - y^2 + 2x + 4y - 8$
- (d) $j(x, y) = -x^2 + y^2 + 2x - 4y + 3$
- (e) $k(x, y) = \sqrt{(x-1)^2 + (y-2)^2}$
- (f) $l(x, y) = -\sqrt{(x-1)^2 + (y-2)^2}$

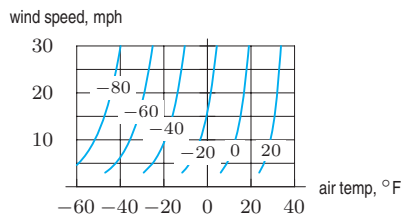
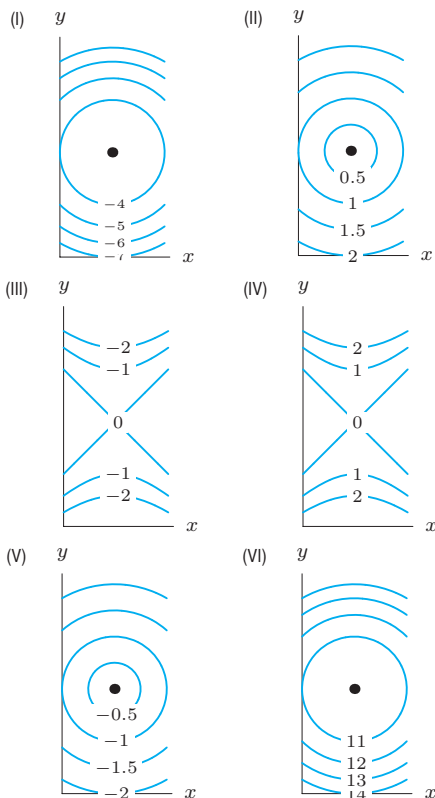
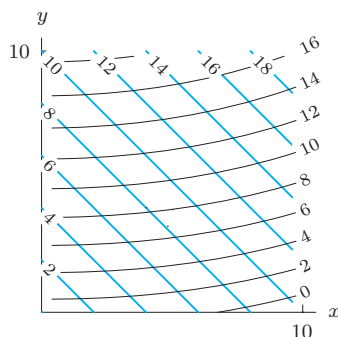


Figure 12.54



23. Figure 12.55 shows contour diagrams of $f(x, y)$ and $g(x, y)$. Sketch the smooth curve with equation $f(x, y) = g(x, y)$.

Figure 12.55: Black: $f(x, y)$. Blue: $g(x, y)$

22. The wind chill tells you how cold it feels as a function of the air temperature and wind speed. Figure 12.54 is a contour diagram of wind chill ($^{\circ}\text{F}$).

- (a) If the wind speed is 15 mph, what temperature feels like -20°F ?
- (b) Estimate the wind chill if the temperature is 0°F and the wind speed is 10 mph.
- (c) Humans are at extreme risk when the wind chill is below -50°F . If the temperature is -20°F , estimate the wind speed at which extreme risk begins.
- (d) If the wind speed is 15 mph and the temperature drops by 20°F , approximately how much colder do you feel?

24. Figure 12.56 shows the level curves of the temperature H in a room near a recently opened window. Label the three level curves with reasonable values of H if the house is in the following locations.

- (a) Minnesota in winter (where winters are harsh).
- (b) San Francisco in winter (where winters are mild).
- (c) Houston in summer (where summers are hot).
- (d) Oregon in summer (where summers are mild).

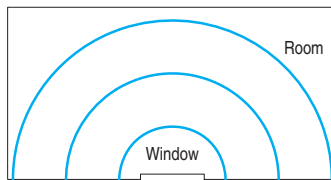


Figure 12.56

25. You are in a room 30 feet long with a heater at one end. In the morning the room is 65°F . You turn on the heater, which quickly warms up to 85°F . Let $H(x, t)$ be the temperature x feet from the heater, t minutes after the heater is turned on. Figure 12.57 shows the contour diagram for H . How warm is it 10 feet from the heater 5 minutes after it was turned on? 10 minutes after it was turned on?

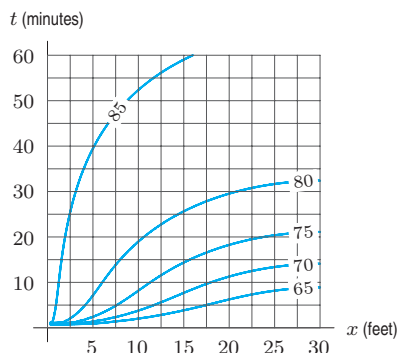


Figure 12.57

26. Using the contour diagram in Figure 12.57, sketch the graphs of the one-variable functions $H(x, 5)$ and $H(x, 20)$. Interpret the two graphs in practical terms, and explain the difference between them.
27. Figure 12.58 shows a contour map of a hill with two paths, A and B.
- On which path, A or B, will you have to climb more steeply?
 - On which path, A or B, will you probably have a better view of the surrounding countryside? (Assume trees do not block your view.)
 - Alongside which path is there more likely to be a stream?

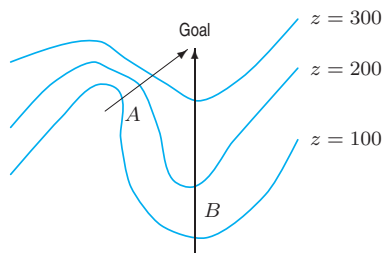


Figure 12.58

28. Figure 12.59 is a contour diagram of the monthly payment on a 5-year car loan as a function of the interest rate and the amount you borrow. The interest rate is 13% and you borrow \$6000 for a used car.
- What is your monthly payment?
 - If interest rates drop to 11%, how much more can you borrow without increasing your monthly payment?
 - Make a table of how much you can borrow, without increasing your monthly payment, as a function of the interest rate.

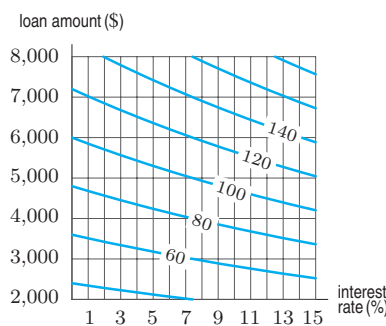


Figure 12.59

29. Hiking on a level trail going due east, you decide to leave the trail and climb toward the mountain on your left. The farther you go along the trail before turning off, the gentler the climb. Sketch a possible topographical map showing the elevation contours.
30. Match the functions (a)–(d) with the shapes of their level curves (I)–(IV). Sketch each contour diagram.
- | | |
|-------------------------|----------------------------|
| (a) $f(x, y) = x^2$ | (b) $f(x, y) = x^2 + 2y^2$ |
| (c) $f(x, y) = y - x^2$ | (d) $f(x, y) = x^2 - y^2$ |
- I. Lines II. Parabolas
III. Hyperbolas IV. Ellipses
31. Figure 12.60 shows the density of the fox population P (in foxes per square kilometer) for southern England. Draw two different cross-sections along a north-south line and two different cross-sections along an east-west line of the population density P .

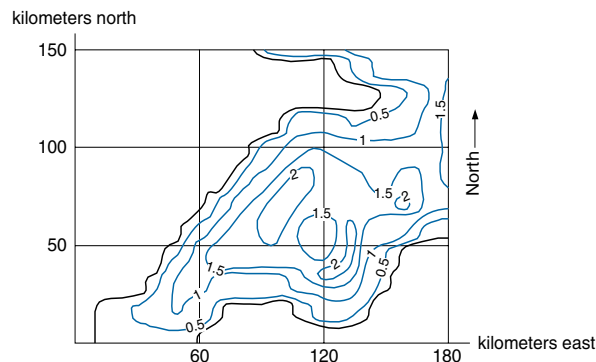


Figure 12.60

32. A manufacturer sells two goods, one at a price of \$3000 a unit and the other at a price of \$12,000 a unit. A quantity q_1 of the first good and q_2 of the second good are sold at a total cost of \$4000 to the manufacturer.
- Express the manufacturer's profit, π , as a function of q_1 and q_2 .
 - Sketch curves of constant profit in the $q_1 q_2$ -plane for $\pi = 10,000$, $\pi = 20,000$, and $\pi = 30,000$ and the break-even curve $\pi = 0$.

33. Match each Cobb-Douglas production function (a)–(c) with a graph in Figure 12.61 and a statement (D)–(G).

(a) $F(L, K) = L^{0.25} K^{0.25}$

(b) $F(L, K) = L^{0.5} K^{0.5}$

(c) $F(L, K) = L^{0.75} K^{0.75}$

(D) Tripling each input triples output.

(E) Quadrupling each input doubles output.

(G) Doubling each input almost triples output.

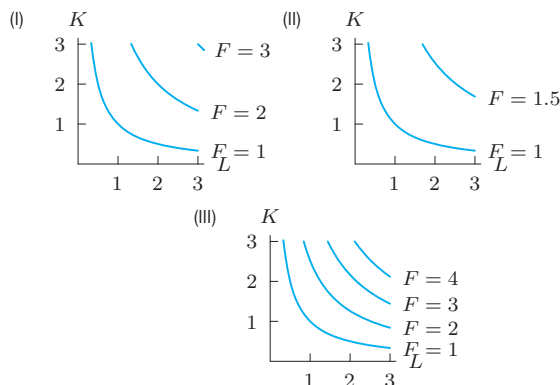


Figure 12.61

34. A Cobb-Douglas production function has the form

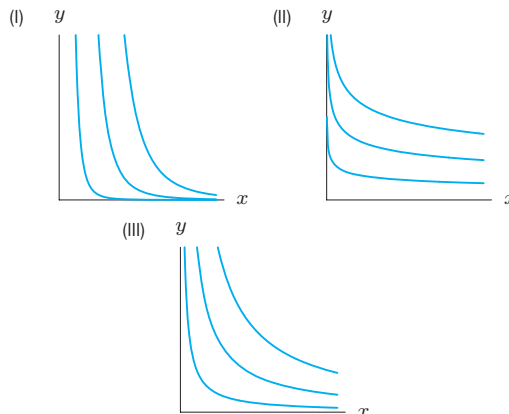
$$P = cL^\alpha K^\beta \quad \text{with } \alpha, \beta > 0.$$

What happens to production if labor and capital are both scaled up? For example, does production double if both labor and capital are doubled? Economists talk about

- *increasing returns to scale* if doubling L and K more than doubles P ,
- *constant returns to scale* if doubling L and K exactly doubles P ,
- *decreasing returns to scale* if doubling L and K less than doubles P .

What conditions on α and β lead to increasing, constant, or decreasing returns to scale?

35. (a) Match $f(x, y) = x^{0.2} y^{0.8}$ and $g(x, y) = x^{0.8} y^{0.2}$ with the level curves in Figures (I) and (II). All scales on the axes are the same.
- (b) Figure (III) shows the level curves of $h(x, y) = x^\alpha y^{1-\alpha}$ for $0 < \alpha < 1$. Find the range of possible values for α . Again, the scales are the same on both axes.



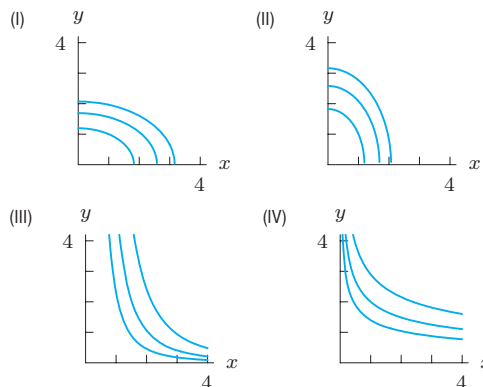
36. Match the functions (a)–(d) with the contour diagrams in Figures I–IV.

(a) $f(x, y) = 0.7 \ln x + 0.3 \ln y$

(b) $g(x, y) = 0.3 \ln x + 0.7 \ln y$

(c) $h(x, y) = 0.3x^2 + 0.7y^2$

(d) $j(x, y) = 0.7x^2 + 0.3y^2$



37. Figure 12.62 is the contour diagram of $f(x, y)$. Sketch the contour diagram of each of the following functions.

(a) $3f(x, y)$

(b) $f(x, y) - 10$

(c) $f(x - 2, y - 2)$

(d) $f(-x, y)$

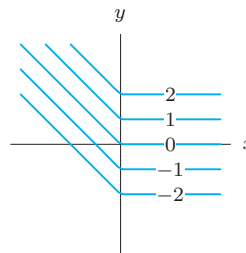


Figure 12.62

38. Figure 12.63 shows part of the contour diagram of $f(x, y)$. Complete the diagram for $x < 0$ if

(a) $f(-x, y) = f(x, y)$ (b) $f(-x, y) = -f(x, y)$

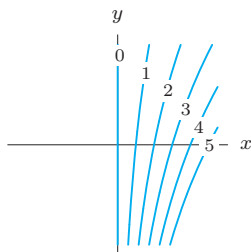


Figure 12.63

39. Let $f(x, y) = x^2 - y^2 = (x - y)(x + y)$. Use the factored form to sketch the contour $f(x, y) = 0$ and to find the regions in the xy -plane where $f(x, y) > 0$ and the regions where $f(x, y) < 0$. Explain how this sketch shows that the graph of $f(x, y)$ is saddle-shaped at the origin.
40. Use Problem 39 to find a formula for a “monkey saddle” surface $z = g(x, y)$ which has three regions with $g(x, y) > 0$ and three with $g(x, y) < 0$.
41. Use the contour diagram for $f(x, t) = \cos t \sin x$ in Figure 12.64 to describe in words the cross-sections of f

with t fixed and the cross-sections of f with x fixed. Explain what you see in terms of the behavior of the string.

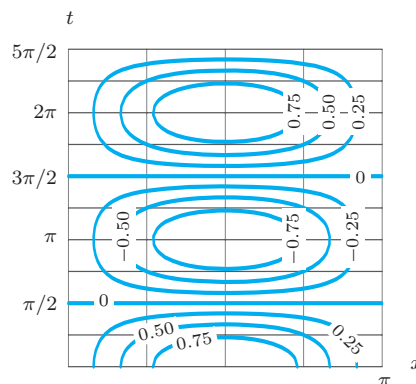


Figure 12.64

42. The power P produced by a windmill is proportional to the square of the diameter d of the windmill and to the cube of the speed v of the wind.⁵
- (a) Write a formula for P as a function of d and v .
- (b) A windmill generates 100 kW of power at a certain wind speed. If a second windmill is built having twice the diameter of the original, what fraction of the original wind speed is needed by the second windmill to produce 100 kW?
- (c) Sketch a contour diagram for P .

Strengthen Your Understanding

In Problems 43–44, explain what is wrong with the statement.

43. A contour diagram for $z = f(x, y)$ is a surface in xyz -space.
44. The functions $f(x, y) = \sqrt{x^2 + y^2}$ and $g(x, y) = x^2 + y^2$ have the same contour diagram.

In Problems 45–46, give an example of:

45. A function $f(x, y)$ whose $z = 10$ contour consists of two or more parallel lines.
46. A function whose contours are all parabolas.

Decide if the statements in Problems 47–51 must be true, might be true, or could not be true. The function $z = f(x, y)$ is defined everywhere.

47. The level curves corresponding to $z = 1$ and $z = -1$ cross at the origin.
48. The level curve $z = 1$ consists of the circle $x^2 + y^2 = 2$ and the circle $x^2 + y^2 = 3$, but no other points.
49. The level curve $z = 1$ consists of two lines which intersect at the origin.
50. If $z = e^{-(x^2 + y^2)}$, there is a level curve for every value of z .

51. If $z = e^{-(x^2 + y^2)}$, there is a level curve through every point (x, y) .

Are the statements in Problems 52–59 true or false? Give reasons for your answer.

52. Two isotherms representing distinct temperatures on a weather map cannot intersect.
53. A weather map can have two isotherms representing the same temperature that do not intersect.
54. The contours of the function $f(x, y) = y^2 + (x - 2)^2$ are either circles or a single point.
55. If the contours of $g(x, y)$ are concentric circles, then the graph of g is a cone.
56. If the contours for $f(x, y)$ get closer together in a certain direction, then f is increasing in that direction.
57. If all of the contours of $f(x, y)$ are parallel lines, then the graph of f is a plane.
58. If the $f = 10$ contour of the function $f(x, y)$ is identical to the $g = 10$ contour of the function $g(x, y)$, then $f(x, y) = g(x, y)$ for all (x, y) .
59. The $f = 5$ contour of the function $f(x, y)$ is identical to the $g = 0$ contour of the function $g(x, y) = f(x, y) - 5$.

⁵From www.ecolo.org/documents/documents_in_english/WindmillFormula.htm, accessed on October 9, 2011.